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MIL-STD-8047
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DEPARTMENT OF DEFENSE STANDARD PRACTICE

QUALITY ASSURANCE AND SHELF-LIFE EXTENSION TESTING FOR PAINTS, SEALERS, ADHESIVES AND RELATED PRODUCTS

AMSC N/A

FSC 80GP

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FOREWORD

1. This Standard is approved for use by all Departments and Agencies of the Department of Defense (DoD).

2. The tables in this Standard are numerous; therefore, they are located after the end of Section 6 but preceding the Appendices.

3. Unless specifically stated, all references to government documents and non-government publications shall refer to the most recent version or revision of that document or publication.

4. References to the Quality Assurance Representative/ Quality Assurance Specialist (QAR/QAS) include Military Service personnel designated to perform these functions.

5. Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Construction and Equipment (C&E), ATTN: FCSB, 700 Robbins Ave. Philadelphia, PA 19111-5098, or emailed to CE_Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://quicksearch.dla.mil/.

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1. SCOPE

1.1 Scope. This Standard provides DoD Policy, general instructions, and minimum procedures to be used by the Military Services and the DLA in performing surveillance quality assurance functions of U. S. Government-owned brushes, paints, sealers, adhesives, and related products worldwide. This Standard also provides Tailored Logistics Support suppliers with minimum procedures for performing shelf-life extension testing management. Copies of this Standard can be obtained through regular channels from ASSIST Online website: https://assist.dla.mil/ or https://quicksearch.dla.mil/.

1.2 Applicability. Quality assurance (QA) is a planned and systematic pattern of all actions necessary to provide confidence that adequate technical requirements are established; products and services conform to established technical requirements; and satisfactory performance is achieved. Quality Surveillance (QS), as used herein and a subset of QA, is the aggregate of measures (stock rotation, sampling, etc.) used to determine and maintain both the quality of product receipts and the Government-owned products to the degree necessary to ensure that such products are suitable for their intended use. A vigilant QS program, implemented by properly trained personnel, is necessary to protect the original product quality and the interests of the Government. Policy and procedures discussed for QS on receiving Government-owned products covered under this standard apply to all Military Service.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information, or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

MMM-A-121	Adhesive, Bonding Vulcanized Synthetic Rubber to Steel
MMM-A-132	Adhesives, Heat Resistant, Airframe Structural, Metal to Metal
MMM-A-134	Adhesive, Epoxy Resin, Metal to Metal Structural Bonding
MMM-A-1617	Adhesive, Rubber Base, General Purpose
TT-C-492	Coating Compound, Paint Antisweat
TT-P-28	Paint, Aluminum, Heat Resisting
TT-P-645	Paint, Alkyd Type, Formula Number 84
TT-P-1757	Primer Coating, ALKYD Base, One Component

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TT-P-1952	Paint, Traffic and Airfield Marking, Waterborne
TT-P-2760	Primer Coating: Polyurethane, Elastomeric, High-Solids
TT-R-248	Remover, Paint and Lacquer, Solvent Type
TT-R-2918	Remover, Paint, No Hazardous Air Pollutants (HAPS)

(Copies of these documents are available online at https://quicksearch.dla.mil.)

COMMERCIAL ITEM DESCRIPTIONS (CID)

A-A-272	Caulking Compounds
A-A-342	Adhesive, Semi-Slid, Stick Form, with Dispenser
A-A-857	Thinner, Dope and Lacquer (Cellulose Nitrate)
A-A-1452	Lacquer (Spraying, Chemical Resistant)
A-A-1558	Paint, Stencil
A-A-2787	Enamel, Aerosol (Low VOC)
A-A-3003	Lacquer, Spraying, Clear and Pigmented for Interior Use
A-A-3007	Thinner: For Phenol-Formaldehyde and Medium Oil and Styrenated Alkyd Paints and Varnishes
A-A-3097	Adhesives, Cyanoacrylate, Rapid Room Temperature-Curing, Solventless
A-A-3108	Tinting Medium, Enamel: Universal Colorants
A-A-3121	Paint, Rubber-Base: for Interior Use (Concrete and Masonry Floors)
A-A-3164	Synthetic Lacquer, Camouflage, Exterior, VOC Complaint
A-A-3165	Lacquer, Gloss, for Aircraft Use
A-A-50535	Adhesive, Starch
A-A-52408	Preservative Coating, Rubber: for Rubber Surface
A-A-52465	Primer Coating, Synthetic, VOC Complaint (for Brake Drums)
A-A-59166	Coating Compound, Nonslip (for Walkways)
A-A-59293	Sealing Compound, Adhesive: Curing (Polysulfide Base)

(Copies of these documents are available online at https://quicksearch.dla.mil.)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-5540	Adhesive, Polychloroprene
MIL-A-21016	Adhesive, Resilient Deck Covering
MIL-A-24179	Adhesive, Flexible Unicellular-Plastic Thermal Insulation
MIL-A-24456	Adhesive for Plastic Vibration-Damping Tile
MIL-A-46106	Adhesive-Sealants, Silicone, RTV, One-Component
MIL-A-46146	Adhesives-Sealants, Silicone, RTV, Noncorrosive (for use with Sensitive Metals and Equipment)
MIL-A-81236	Adhesive, Epoxy Resin with Polyamide Curing Agent
MIL-C-8514	Coating Compound, Metal Retreatment, Resin-Acid (ASG)
MIL-C-10578	Corrosion Removing and Metal Conditioning Compound (Phosphoric Acid Base)
MIL-DTL-450	Coating Compound, Bituminous Solvent Type, Black (for Ammunition)

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MIL-DTL-1115	Enamel, Interior, Alkyd, White (Formula No. 30)
MIL-DTL-3316	Adhesives, Fire-Resistant, Thermal Insulation
MIL-DTL-15090	Enamel, Equipment, Light Gray (Navy Formula No.111)
MIL-DTL-24441	Paint, Epoxy-Polyamide, General Specification for
MIL-DTL-24607	Enamel, Interior, Nonflaming (Dry), Chlorinated Alkyd Resin, Semigloss
MIL-DTL-24631	Paint, Camouflage for Submarines, General Specification for (Metric)
MIL-DTL-53022	Primer, Epoxy Coating, Corrosion Inhibiting Lead and Chromate Free
MIL-DTL-53030	Primer Coating, Epoxy, Water Based, Lead and Chromate Free
MIL-DTL-53039	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
MIL-DTL-53084	Primer, Cathodic Electrodeposition, Chemical Agent Resistant
MIL-DTL-64159	Camouflage Coating, Water Dispersible Aliphatic Polyurethane, Chemical Agent Resistant
MIL-DTL-81706	Chemical Conversion Materials for Coating Aluminum and Aluminum Alloys
MIL-DTL-81772	Thinner, Aircraft Coating
MIL-E-22118	Enamel, Electrical-Insulating
MIL-M-24041	Molding and Potting Compound, Chemically Cured, Polyurethane
MIL-P-21035	Paint High Zinc Dust Content, Galvanizing Repair (Metric)
MIL-PRF-3043	Resin Coating, Permanent, for Engine Components and Metal Parts
MIL-PRF-6799	Coatings, Sprayable, Strippable, Protective, Water Emulsion
MIL-PRF-8116	Putty, General Purpose, Non-Hardening
MIL-PRF-8516	Sealing Compound, Synthetic Rubber, Electric Connectors and Electric Systems, Chemically Cured
MIL-PRF-9117	Adhesive; Sealing, For Aromatic Fuel Cells and General Repair
MIL-PRF-14105	Paint, Heat-Resisting (for Steel Surfaces)
MIL-PRF-16173	Corrosion Preventive Compound, Solvent Cutback, Cold-Application
MIL-PRF-19565	Coating Compound, Thermal Insulation, Fire-and Water-Resistant, Vapor-Barrier
MIL-PRF-22750	Coating, Epoxy, High-Solids
MIL-PRF-23236	Coating Systems for Ship Structures
MIL-PRF-23377	Primer Coatings: Epoxy, High-Solids
MIL-PRF-23586	Sealing Compound (with Accelerator), Silicone Rubber, Electrical
MIL-PRF-24176	Cement, Epoxy, Metal Repair and Hull Smoothing (Metric)
MIL-PRF-24596	Coating Compounds, Nonflaming, Fire-Resistant
MIL-PRF-24635	Coating Systems, Weather-Resistant, Exterior Use
MIL-PRF-24647	Paint System, Anticorrosive and Antifouling, Ship Hull
MIL-PRF-32171	Deck Coatings, High Durability
MIL-PRF-32348	Powder Coating, Camouflage Chemical Agent Resistant Systems
MIL-PRF-32550	Metal-Rich Primer
MIL-PRF-81352	Coatings, Aircraft Touch-Up
MIL-PRF-81733	Sealing and Coating Compound, Corrosion Inhibitive

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MIL-PRF-83936	Remover, Paint, Tank Type: For Aircraft Wheels, Landing Gear Components, and Other Aircraft and Support Equipment
MIL-PRF-85285	Coating: Polyurethane, Aircraft and Support Equipment
MIL-PRF-85582	Primer Coatings: Epoxy, Waterborne
MIL-R-9300	Resin, Epoxy, Low-Pressure Laminating
MIL-R-81294	Remover, Paint, Epoxy, Polysulfide, and Polyurethane Systems
MIL-S-11030	Sealing Compound, Single-Component, Non-Curing, Polysulfide Base
MIL-S-15204	Sealing Compound, Joint and Thread, High Temperature
MIL-S-45180	Sealing Compound, Gasket, Hydrocarbon Fluid and Water Resistant
MIL-S-85334	Sealing Compound, Noncuring, Low Consistency, Silicone, Groove Injection, for Integral Fuel Tanks

(Copies of these documents are available online at https://quicksearch.dla.mil.)

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-831 Preparation of Test Reports

(Copies of these documents are available online at https://quicksearch.dla.mil.)

DEPARTMENT OF DEFENSE PUBLICATIONS

DODM 4140.27, Vol 1 DoD Shelf-Life Management Program: Program Administration
DODM 4140.27, Vol 2 DoD Shelf-Life Management Program: Materiel Quality Control Storage Standards

(Copies of these documents are available online at https://www.esd.whs.mil/dd/.)

JOINT SERVICE PUBLICATIONS

DLAR (JP) 4155.37 DoD Shelf-Life Materiel Quality Control Storage Standards

(Copy of this document is available online at https://www.dla.mil/.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

US GOVERNMENT FORMS

DD FORM 1222 Request for and Results of Tests

(Copies of these documents are available online at https://www.esd.whs.mil/dd/.)

US GOVERNMENT STANDARD FORM

SF 368 Product Quality Deficiency Report

(Copies of this document are available online at https://www.gsa.gov/.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4	Sampling Procedures and Tables for Inspection by Attributes
ANSI/NCSL Z540.3	Requirements for the Calibration of Measuring and Test Equipment

(Copy of this document is available online from https://www.ansi.org.)

ASTM INTERNATIONAL

ASTM D3925	Standard Test Method for Sampling Liquid Paints and Related Pigmented Coatings
ASTM D5363	Standard Specification for Anaerobic Single-Component Adhesives (AN)
ASTM E300	Standard Practice for Sampling Industrial Chemicals

(Copies of these documents are available online from https://www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 9001	Quality Management Systems Requirements
ISO 10012	Measurement Management Systems-Requirements for Measurement Processes and Measuring Equipment
ISO/IEC 17025	General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of these documents are available online from https://www.iso.org/iso/catalogue ics.)

MASTER PAINTERS INSTITUTE

MPI 1	Aluminum Paint
MPI 2	Aluminum Paint, Heat Resistant (Up To 427ºC - 800ºF)
MPI 10	Latex, Exterior Flat (MPI Gloss Level 1)
MPI 10RR	Latex, Recycled (Remanufactured), Exterior Flat (G1)
MPI 11	Latex, Exterior, Semi-Gloss (MPI Gloss Level 5)
MPI 13	Stain, Exterior, Solvent Based, Semi-Transparent
MPI 15RR	Latex, Recycled (Remanufactured), Exterior, Low Sheen (G3-4)
MPI 27	Floor Enamel, Alkyd, Gloss (MPI Gloss Level 6)
MPI 31	Varnish, Polyurethane, Moisture Cured, Gloss (MPI Gloss Level 6)
MPI 32	Traffic Marking Paint, S.B. (Solvent-Based)
MPI 94	Alkyd, Exterior Semi-Gloss (MPI Gloss Level 5)
MPI 99	Sealer, Water Based, For Concrete Floors
MPI 119	Latex, Exterior, Gloss (MPI Gloss Level 6)
MPI 138	Latex, Interior, High Performance, ('velvet-like' - MPI Gloss Level 2)
MPI 141	Latex, Interior, High Performance, Semi-Gloss (MPI Gloss Level 5)

(Copies of these documents are available online from https://mpiinfo.org/store/)

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SAE INTERNATIONAL

SAE AMS3265	(R) Sealing Compound, Polysulfide (T) Rubber, Fuel Resistant, Non-Chromated Corrosion Inhibiting For Intermittent Use to 360 °F (182 °C)
SAE AMS3276	Sealing Compound, Integral Fuel Tanks and General Purpose, Intermittent Use to 360 °F (182 °C)
SAE AMS3277	(R) Sealing Compound, Polythioether Rubber Fast Curing for Integral Fuel Tanks and General Purpose, Intermittent Use to 360 °F (182 °C)
SAE ASM3284	(R) Sealing Compound, Low Adhesion, for Removable Panels and Fuel Tank Inspection Plates
SAE AMS3374	Sealing Compound Aircraft Firewall, Silicone
SAE AMS-C-83231	Coatings, Polyurethane, Rain Erosion Resistant for Exterior Aircraft and Missile Plastic Parts
SAE AMS-S-4383	Sealing Compound, Topcoat, Fuel Tank Buna-N Type
SAE AMS-S-7124	Sealing Compound, Polysulfide, Accelerator Required, For Aircraft Structures
SAE AMS-S-8802	(R) Sealing Compound, Fuel Resistant, Integral Fuel Tanks and Fuel Cell Cavities

(Copies of these documents are available online at https://www.sae.org.)

2.4 Order of Precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Exception: shelf-life extension test requirements in product specifications will take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS.

3.1 Acquisition Management Systems Control (AMSC) number. A control number assigned by the cognizant data management approval authority listed in Standardization Directory (SD-1), which indicates that a data item description (DID) or a defense specification or standard that cites DIDs has been cleared for use by the DoD.

3.2 Acquisition Streamlining and Standardization Information System (ASSIST). The official database containing information about standardization documents used in the DoD. ASSIST also provides electronic access to government documents included in the database over the Internet. ASSIST can be accessed at https://assist.dla.mil/ or https://quicksearch.dla.mil.

3.3 Analysis. An element of verification that uses a detailed examination by established technical or mathematical models or simulations, algorithms, charts, graphs, circuit diagrams, or other scientific principles and procedures to provide evidence that stated requirements were met.

3.4 Appearance. The visual characteristics of an item such as color, clarity evidence of

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stratification and contamination that may be observed by visual examination of the sample.

3.5 Batch or Lot. A specific quantity of product that is produced, processed or utilized as a single unit and tested to meet test criteria and specifications.

3.6 Batch number or Lot number. A unique number that is assigned by the manufacturer that allows traceability to specified quantity of product, history of its production as well as relevant issues of control and manufacturing.

3.7 Calibration. The comparison of measured values of the significant parameters with values indicated by a set of reference standards with known accuracy.

3.8 Certificate of Analysis (CoA). A CoA is issued by a laboratory that is preferably not from the originating manufacture that describes the product.

3.9 Class. An additional categorization of differences in characteristics, other than that afforded by type classification. It does not constitute a difference in quality or grade, but is for specific, equally important uses, and is usually designated by Arabic numerals, such as, "class 1" or "class 2."

3.10 Commingling. The mixing of two or more products of different ownership, type, or grade into a harmonious whole.

3.11 Contaminant. A foreign substance in a product.

3.12 Contaminated product. A product containing foreign matter such as dust, rust, water, or emulsions to the extent it changes one or more characteristics of the product.

3.13 Defense specification. A specification developed under the consensus procedures of the Defense Standardization Program that is used on multiple programs or applications and is included in the ASSIST database. Defense specification establishes requirements for military-unique products or commercial items that must be substantively modified to include military-unique requirements.

3.14 Design activity. The organization that prepares, or has another prepare for it, a program unique specification.

3.15 Detail specification. A specification that states design requirements, such as materials to be used, how a requirement is to be achieved, or how an item is to be fabricated or constructed. A specification that contains both performance and detail requirements is still considered a detail specification. Both defense specifications and program-unique specifications may be designated as a detail specification.

3.16 Deteriorated product. A product in which one or more characteristics have changed to a level of quality outside the limits of the applicable specification (lower quality than original specifications).

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3.17 Dimensional-life. The term dimensional-life is newly introduced in AS23053 to relate to the dimensional stability instead of stability of the property stability often associated or referred to as the shelf-life.

3.18 DoD-approved laboratory. A laboratory that has been certified by an entity acting under the purview of the DoD Shelf-Life Program office as authorized to perform shelf-life extension testing of DoD materiel. DoD-approved laboratories are listed on SLES and undergo recertification on a regular recurring basis to maintain current DoD-approved status.

3.19 Equivalent tests. Test methods that provide analogous results and fully correlate with standard ASTM methods, but have not yet been formally accepted by ASTM. These test methods, such as SAE international, FED-STD-791, or methods spelled out in the specification, have been found to provide test results essentially identical to those produced by ASTM testing methods.

3.20 Examination. An element of verification that is generally nondestructive and typically includes the use of sight, hearing, smell, touch, and taste; simple physical manipulation; and mechanical and electrical gauging and measurement.

3.21 Federal Supply Class (FSC) codes. FSC codes are used to group products into logical families for management purposes. The four-digit fields are used to group standardization documents and their products into logical families.

3.22 Inspect. To examine critically especially to detect flaws, errors, etc.

3.23 Item Type Storage Code (ITSC). A code that indicates the type of storage space and environment conditions to be maintained for an item during storage and shipment (ITSC also referred to as TOS).

3.24 Materiel Quality Control Storage Standard (MQCSS). Instructions for the inspection, testing, and restoration of items in storage, which encompass requirements including type of storage environment, preservation, packaging, exercising, and inspection/testing criteria with time-phasing to determine the serviceability status of materiel in storage and the degree of degradation that has occurred.

3.25 Non-Government standard. A national or international standardization document developed by a private sector association, organization, or technical society that plans, develops, establishes, or coordinates standards, specifications, handbooks, or related documents. The term does not include standards of individual companies. Non-Government standards adopted by the DoD are listed in the ASSIST database.

3.26 Off-Specification product. A product which fails to meet one or more of the physical, chemical, or performance requirements of the specification.

3.27 Packaging. The processes and procedures used to protect material from deterioration, damage, or both during shipment and storage. It includes cleaning, drying, preserving, packing, marking, and unitization.

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3.28 Performance specification. A specification that states requirements in terms of the required results with criteria for verifying compliance, but without stating the methods for achieving the required results. A performance specification defines the functional requirements for the item, the environment in which it must operate, and interface and interchangeability characteristics. Both defense specifications and program-unique specifications may be designated as a performance specification.

3.29 Product Quality Deficiency Report (PQDR) SF 368. The PQDR is a report submitted by a customer or depot to report an off-specification product condition or an operational issue where the product is suspected of causing the problem. It is used to identify, report, and resolve conditions negatively impacting the war fighter. The PQDR process provides timely quality feedback to activities responsible for design, development, purchasing, production, supply, maintenance, contract administration, and other functions so that action can be initiated to determine cause, take corrective action, and prevent recurrence. Integrates deficiency analysis and resolution processes to identify root cause and prevent or mitigate recurrence within acquisition, quality, systems engineering, and overall life cycle management plans. It allows originating activities a means to obtain cost credit, replacement, and/or contractual remedy for procurement related quality deficiencies resulting from poor workmanship, nonconformance to applicable specifications, drawings, standards, processes or other technical requirements.

3.30 Qualified Product Database (QPD). This database consists of the officially approved electronic QPLs. Only those electronic QPLs published in the QPD are the official source for qualified products. The link to the QPD webpage is through Assist Online.

3.31 Qualified Product List (QPL). A list of products or families of products that have successfully completed the formal qualification process (including all specified periodic tests) that examines, tests and verifies that a specific product design meets all applicable specification requirements. The link to the QPL webpage is through Assist Online.

3.32 Quality. The composite of materiel attributes including performance, features and characteristics of a product, or service to satisfy a given requirement.

3.33 Quality Assurance (QA). A system of activities, the purpose of which is to provide to the producer and user of a product, measurement or service the assurance that it meets the defined standards of quality with a stated level of confidence and includes quality planning and quality control. A planned and systematic pattern of all actions necessary to ensure that adequate technical requirements are established; that products, quantity accountability, and services conform to these established technical requirements; and that satisfactory performance is achieved. It includes: Quality Planning during specification development and review; Quality Support to contracting and acquisition teams; Quality Oversight of product and service providers to assure compliance to contracts and agreements; Quality Control (QC) operations for products and services incoming or in the Government supply chain; and, Quantity Measurement and control activities. Generically the term QA refers to all processes and procedures encompassing Quality planning/development, QA, QS, and QC.

3.34 Quality Assurance Representative/Quality Assurance Specialist (QAR/QAS). An organizational title assigned to the individual responsible for Government contract QA functions.

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QARs have cognizance over the procurement of product and/or services at contractor facilities. The Military Services may use other organizational titles for their personnel performing QA and QS functions.

3.35 Quality Status List (QSL). The DoD recognized centralized repository of test results of Type II shelf-life item samples that have been tested by DoD-approved laboratories. The QSL listing of accumulated test results in which the laboratory testing of Type II shelf-life item samples have been accomplished by DoD-approved laboratories, and can be used as an authority for extending inventories with the same identification as long as MQCSS requirements have been adhered to.

3.36 Quality Surveillance (QS). A subset of QA encompassing the program of inspections, sampling, testing, quantity measurement and control, and documentation established to monitor the quality/quantity of product being received, stored and issued within the Government supply chain.

3.37 Repeatability. Difference between successive test results, obtained by the same operator with same apparatus under constant operating conditions on identical test material would, in the long term and correct operation of the test method, exceed the values given only in one case in twenty.

3.38 Representative Sample. A portion extracted from a larger volume that adequately replicates the larger volume (group) characteristics and quality.

3.39 Reproducibility. Quantitative expression of the random error associated with operators working in different laboratories, each obtaining single results on identical test material when applying the same method and the same previously agreed acceptance criteria.

3.40 Sample. A portion extracted from a total volume which may or may not contain constituents in the same proportions that are present in that total volume.

3.41 Sample tag. DD Form 1222, Request for and Results of Tests form used for identification of samples and their origin.

3.42 Shelf-Life Extension System (SLES). The web-based system is the central DoD data repository for both the MQCSS information and QSL DoD-approved laboratory test results. The MQCSS system is the authority for shelf-life extension when only visual inspection is required. The MQCSS system and QSL system are the authority for shelf-life extension when visual inspection and laboratory or machine testing are required. Note shelf-life extension is only applicable to Type II (extendable) items. SLES is at https://www.shelflife.dla.mil/.

3.43 Shelf-Life, Type I. A definite non-extendable period of shelf-life.

3.44 Shelf-Life, Type II. An assigned shelf-life period that may be extended after successful completion of visual inspection, certified laboratory test, restorative action, or a combination of these measures.

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3.45 Specification. A document prepared to support acquisition that describes essential technical requirements for material and the criteria for determining whether those requirements are met. Specifications can be classified as Federal, Military, Commercial, or Voluntary Standard. Specifications can be categorized as functional, design, or a performance specification.

3.46 Specification limits. Boundaries or parameters that define acceptable performance for a characteristic expressed as a target maximum or minimum limit, or both an upper and lower limit (range).

3.47 Testing. Analytical techniques using instruments, equipment, and other methods to predict paints, sealers, adhesives and related products test values using compositional data typically determined by chemistry methods.

3.48 Thief. A sampling apparatus designed so a liquid sample can be obtained from any specified point in the container.

3.49 Tolerance. Allowed variations within a specified standard.

3.50 Type of Storage (TOS). See Item Type Storage Code.

4. GENERAL REQUIREMENTS

4.1 Personnel competency. Personnel responsible for handling product shall be thoroughly trained and fully qualified to perform assigned responsibilities. They shall be aware of the hazards in handling products, as well as the applicable safety and operating procedures.

4.2 Equivalent tests. The use of alternate test methods, such as SAE International, FED-STD-791, or methods spelled out in the specification, is allowed provided that: the test results are presented in the format required in the specification; the test device has a demonstrated reliability and repeatability equal or better than called for by the ASTM method; and the device has been approved for use by the military services.

4.3 Calibrating test equipment. All laboratories shall ensure calibration of testing and measuring equipment is up to date and is to the accuracy necessary to ensure the equipment is within allowable tolerance limits. Calibration records shall include “as received/as left” values.

4.4 Specifications. Each product testing laboratory shall maintain an up-to-date file of specifications (both Military and non-Government specifications and standards). It is not practical to include complete specification limits in this Standard, as specifications are subject to change with variations in product availability and technical developments. Limits cited in this Standard are for internal government use, not for procurement.

4.5 Safety precautions. Throughout this Standard, there are general safety precautions and instructions that apply to product handling and laboratory operations to ensure personal safety/health and the protection of Government property. Occupational Safety and Health

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Administration (OSHA), Department of Labor, and standard commercial safety practices, such as Safety Data Sheets, shall be observed.

4.6 Compliance with regulations and laws. Many shelf-life products are hazardous materials and are regulated as such. Users of this Standard must be aware of regulations and laws governing the products that they are handling. In the event of a conflict between this Standard and a law or regulation, the law or regulation takes precedence.

4.7 Toxic chemicals, hazardous substances, and ozone-depleting chemicals. The use of toxic chemicals, hazardous substances, or ozone-depleting chemicals shall be avoided, if feasible. If a toxic chemical, hazardous substance, or ozone-depleting chemical must be used. The Environmental Protection Agency (EPA) maintains an online list of toxic chemicals and hazardous substances at http://www.epa.gov/emergencies/tools.htm#lol and should be consulted.

4.8 Test methods. Laboratories shall maintain an up-to-date record of all test methods performed by the laboratory.

5. DETAILED REQUIREMENTS

5.1 Receipt of products. Products should be inspected upon arrival for correct identification, content, packaging and damage. Containers should be in good condition upon arrival, and the supplier should supply proper paper work. Store according to the Item Type Storage Code (ITSC/TOS) or in lieu of the ITSC using the specification or manufacturer requirements. In the situation where storage must be outside, containers must be off the ground on pallets or dunnage and covered with tarpaulins for protection from the elements. The proper packaging, storage, and stock rotation (first in – first out) will help to minimize losses due to deterioration.

5.2 Sampling. All samples shall be taken as prescribed by the Materiel Quality Control Storage Standard (MQCSS), product specifications, and ANSI/ASQ Z1.4.

5.3 Precautions. The precautions required to ensure a representative sample are many and depend on type of product being sampled, the type of container from which it is drawn and the sampling procedures employed. Each procedure is suitable for sampling a specific product under definite storage, transportation and container conditions. Warning: “All safety instructions shall be strictly observed.”

5.4 Personnel to conduct sampling. Because improperly taken samples can completely invalidate a test, only trained and experienced personnel shall be assigned to sample the products. This cannot be overstressed; no amount of laboratory work will give reliable data on a product if the sample is not a true representation of that product.

5.5 Responsibility. This Standard shall in no way alter any assigned responsibility of the various activities outside the continental United States for submitting special samples to a designated laboratory or as directed by the cognizant headquarters.

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5.6 Types of samples. A sample is a portion of a product, batch, or lot taken which represents that entire group. The various types of samples are as follows:

5.6.1 Tube or thief sample is one obtained with a sampling tube or special thief either as a core or spot sample from a specified point in the container.

5.6.2 Batch or lot sample is one obtained from a collection of units of package products.

5.7 Sampling apparatus, containers, and procedures.

a. Approved type samplers shall be used as specified by ASTM procedures. All sampling apparatus and containers shall be thoroughly clean and dry, and special care shall be taken so no lint or fibrous material remains in or on them.

b. The Sampling Apparatus must be cleaned as specified by ASTM procedures. When no procedure exist, the apparatus must be flushed three times with the sample. Each flush shall be 10 to 20% of the volume of the sampling apparatus. For each flush, the apparatus shall be closed and shaken for 10 seconds and product replaced for the next flush. Sampling apparatus shall be cleaned immediately after use and stored so it shall remain clean until next use.

c. Containers such as drums shall be sampled with a thief. In sampling drums and cans, care shall be taken to remove all foreign matter from the area near the enclosure before the plug is removed.

d. Close all sample containers tightly, immediately after taking the sample. Do not add sealing wax, paraffin, rubber gaskets, pressure sensitive tapes, or similar material to seal containers, unless the container is designed to use those for closure.

e. Light sample containers shall be adequately crated to withstand shipment. To prevent leakage caused by thermal expansion of the product, do not fill any sample container above 80% volume capacity.

5.8 Size of samples.

a. Except for liquid units of issue greater than 1-gallon and semi-solids greater than a 6.5-pound can or container, all samples shall be submitted in the original, unopened container. When instructed to take a sample, the sample size shall be as follows:

i. Liquid

	Unit of Issue	Sample Size
1.	Less than 1 quart	Approximately 2 quarts- in the individual original containers
2.	1-quart can	2 quarts

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3. 1-gallon can	1-gallon can
4. Any unit of issue larger than 1 gallon	1 gallon, sampled and containerized per 5.7 above

ii. Sprayable Products

Unit of Issue	Sample Size
1. 1- can	2 cans
2. Any unit larger than 1 can	2 cans

iii. Semi-solid

Unit of Issue	Sample Size
1. Less than 1 pound	2 pounds, in the individual original containers
2. 1-pound can	two 1-pound cans
3. 5 pound can or container	One 5-pound can or container
4. 6.5-pound can or container	One 6.5-pound can or container
5. Any unit of issue larger than 6.5 pounds	One pound, sampled and containerized per 5.7 above

b. For container sizes not listed in the information above, contact the appropriate product focal point or a DoD-approved Laboratory for guidance.

5.9 Identification of samples. Identify each sample container immediately after sampling by securely attaching or collocating a completed DD Form 1222, Request for and Results of Tests.

Information on the form shall include the following:

a. To – the testing laboratory

b. From – Storage location

c. Manufacturing plant name and address

d. Contract number

e. End item (as applicable)

f. National Stock Number

g. Sample number – number of samples submitted

h. Quantity Submitted

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i. Lot or batch number (as applicable)

j. Reason for submittal – Shelf-life extension testing

k. Date submitted

l. Material to be tested

m. Quantity represented

n. Date of manufacturer

o. Specification & amendment, drawing or part number

p. Shipment method

q. Date sampled and submitted by

r. MQCSS/QSL Review – check box

s. Remarks and/or special instructions and/or waivers

t. Send report of test to – email address

5.10 Testing. The quality surveillance segment (testing) presented in this section is the minimum essential to sound management of Government-owned properties. Before testing, review the QSL to verify the product lot or batch has not already been tested.

5.10.1 Testing frequency and extension periods. For current testing frequency, extension period, and maximum number of extension requirements refer to the Materiel Quality Control Storage Standards (MQCSS) at https://www.shelflife.dla.mil/.

5.10.2 Test methods. All laboratory tests shall be conducted in accordance with the method prescribed in the specification covering the product, except any special or modified method outlined in this Standard which shall be used in lieu of the specification method when products are evaluated within the scope of this Standard.

5.10.3 Specification receipt limits. Specification receipt limits are absolute. Multiple tests may be performed and if these tests do not differ from each other by more than the amount specified for the reproducibility of the method, the results may be averaged to determine compliance with the specification limits.

5.10.4 Minimum testing. Tables I - V are mandatory and outline the minimum sampling and testing requirements considered necessary for determining the quality of related products to extend the shelf-life. It covers the conditions under which a sample is taken, the type of sample and the types of tests required to determine whether the quality is within acceptable limits.

5.10.5 Alternate test methods. The use of alternate test methods is allowed, provided that: test results are presented in the format required in the specification; the test device has a demonstrated reliability and repeatability equal to or better than that called for by the ASTM and

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the device has been approved for use by the military services. The types of alternate tests are listed below:

a. Equivalent tests are test methods that provide analogous results and fully correlate with standard ASTM methods, but, have not yet been formally accepted by ASTM. These test methods, such as SAE International, FED-STD-791, or methods spelled out in the specification, have been found to provide test results that will be essentially identical to those results produced by ASTM testing methodologies.

b. Predictive testing involves the use of instrumental and other types of analytical techniques to predict test values using compositional data that typically is determined by standard or wet chemistry methods.

5.10.6 Calibrating test equipment. All laboratories shall calibrate testing and measuring equipment to the accuracy necessary to ensure the equipment is within allowable tolerance limits. ISO 10012 or service defined requirements shall be used. See ANSI/NCSL Z540.3 and ISO/IEC 17025 as applicable.

5.11 Disposition procedures. Prior to submission of samples for testing or prior to reclassification of condition codes, the storage activity/customer/laboratory shall consult the DoD QSL to determine the status of the particular batch/lot number under a specific contract.

a. If the QSL indicates condition code “A”, then the stock shall be updated accordingly.

b. If the QSL indicates condition code “H”, then the stock shall be disposed of through DLA Disposition Services, formerly Defense Reauthorization Marketing Office (DRMO), in accordance with local procedures.

c. If the item is not listed in the QSL, then the appropriate focal point shall be contacted.

5.11.1 Service and DLA responsibilities. Review the DoD Shelf-Life Program website at https://www.shelflife.dla.mil/ for the Service and DLA Shelf-Life Representative under Points of Contact.

5.12 Packaging and storage products. The care and preservation of supplies in a ready-for-issue condition, from supplier to user, is an important responsibility of the military services. Strict adherence to storage temperatures for all products is required. The appropriate military activity shall prescribe the procedures and establish the requirements in each phase of the storage program. The organization owning these products has an inherent obligation to ensure that such storage conditions are enforced. These shall be predicated on the type of item, type of storage, anticipated length of storage, probable end use and other factors.

5.12.1 Documentation. Care of supplies in storage is a program of such magnitude that detailed procedures cannot be included in this Standard. Reference shall be made to appropriate departmental publications. Pertinent highlights are cited in the following paragraphs.

5.12.2 Drum storage. Except in an emergency, containers shall not be stored in direct contact with the ground. Drums shall be stored on their sides on dunnage with proper blocking

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and bracing. Bungs shall be in a horizontal position so leaks may be detected and/or eliminated. Drums shall never be stored vertically outdoors as water shall collect on drum heads, seep through bungs and contaminate the product.

5.12.3 Separate storage. For identification purposes, different products shall be stored separately. Oldest stocks shall be used first.

5.12.4 Stock rotation. Where feasible, products opened for spot checking or storage control testing shall be used as soon as possible. When this cannot be done, the containers shall be reclosed tightly, marked as having been previously opened and be included in the next issue if possible. To minimize deterioration of a product due to age, excessive corrosion of containers, and/or deterioration of packing and markings, (excluding other quality considerations), the oldest products shall be issued first.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The purpose of this Standard is to establish common requirements for extension of product shelf-life. This Standard is military unique because it covers internal government procedures for the handling and storage of government-owned products under environmental conditions not generally encountered in the commercial sector.

6.2 Subject term (key word) listing

Alkyd
Corrosion
Enamel
Lacquer
Latex
Pigmented coatings
Polyurethane
Powder coating
Primer coating
Stain
Thinner
Tinting
Varnish

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TABLE I. Tests for FSC 8010 Paint, Dope, Varnish, and Related Products.

CHARACTERISTICS	SPECIFICATION					
	A-A-1452	A-A-1558	A-A-2787	A-A-3003	A-A-3108	A-A-3121
Condition in Container	X	X	X	X	X	X
Solids Content						
Viscosity/Consistency		X		X	X	
Drying/ Cure Time	X	X	X	X		X
Application/Spray Properties	X	X	X	X	X	X
Pot Life						
Adhesion[1]	X	X		X		X
Gloss	X	X	X	X		X
Color	X	X	X	X	X	X
Appearance of Applied Coating	X	X	X	X		
Mixing	X	X	X	X	X	X
Self-Lifting			X			

TABLE I. Tests for FSC 8010 Paint, Dope, Varnish, and Related Products – Continued.

CHARACTERISTICS	SPECIFICATION					
	A-A-3164	A-A-3165	A-A-52465	A-A-59166	AMS-C-83231	MIL-DTL-1115
Condition in Container	X	X	X	X	X	X
Solids Content				X	X	
Viscosity/Consistency	X	X			X	X
Drying/ Cure Time	X	X	X	X	X	X
Application/Spray Properties	X	X	X	X	X	X
Pot Life					X	
Adhesion[1]	X	X	X			X
Gloss	X	X				X
Color	X	X		X		X
Appearance of Applied Coating	X	X	X	X	X	X
Mixing	X	X	X	X	X	X
Self-Lifting	X	X				

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TABLE I. Tests for FSC 8010 Paint, Dope, Varnish, and Related Products – Continued.

CHARACTERISTICS	SPECIFICATION					
	MIL-DTL-15090	MIL-DTL-24441	MIL-DTL-24607	MIL-DTL-24631	MIL-DTL-53022	MIL-DTL-53030
Condition in Container	X	X	X	X	X	X
Solids Content						
Viscosity/Consistency	X	X	X	X	X	
Drying/ Cure Time	X	X	X	X	X	X
Application/Spray Properties	X	X	X	X	X	X
Pot Life		X		X		X
Adhesion[1]	X	X	X	X	X	X
Gloss	X	X	X	X	X	X
Color	X	X	X	X	X	
Appearance of Applied Coating	X	X	X	X	X	X
Mixing	X	X	X	X	X	X
Self-Lifting						

TABLE I. Tests for FSC 8010 Paint, Dope, Varnish, and Related Products – Continued.

CHARACTERISTICS	SPECIFICATION					
	MIL-E-22118	MIL-P-21035	MIL-PRF-14105	MIL-PRF-22750	MIL-PRF-23236	MIL-PRF-23377
Condition in Container	X	X	X	X	X	X
Solids Content						
Viscosity/Consistency	X		X	X	X	X
Drying/ Cure Time	X	X	X	X	X	X
Application/Spray Properties	X	X	X	X	X	X
Pot Life					X	X
Adhesion[1]	X		X	X	X	X
Gloss			X	X	X	
Color			X	X	X	X
Appearance of Applied Coating	X	X	X	X	X	X
Mixing	X	X	X	X	X	X
Self-Lifting						

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TABLE I. Tests for FSC 8010 Paint, Dope, Varnish, and Related Products – Continued.

CHARACTERISTICS	SPECIFICATION					
	MIL-PRF-24596	MIL-PRF-24635	MIL-PRF-24647	*MIL-PRF-26915	MIL-PRF-81352	MIL-PRF-85285
Condition in Container	X	X	X	X	X	X
Solids Content						
Viscosity/Consistency	X	X	X		X	X
Drying/ Cure Time	X	X	X	X	X	X
Application/Spray Properties	X	X	X	X	X	X
Pot Life			X	X		X
Adhesion[1]	X	X		X	X	X
Gloss	X	X			X	X
Color	X	X	X	X	X	X
Appearance of Applied Coating	X	X	X	X	X	X
Mixing	X	X	X	X	X	X
Self-Lifting						

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TABLE I. Tests for FSC 8010 Paint, Dope, Varnish, and Related Products – Continued.

CHARACTERISTICS	SPECIFICATION						
	MIL-PRF-85582	MPI 1	MPI 10	MPI 10RR	MPI 11	MPI 119	MPI 13
Condition in Container	X	X	X	X	X	X	X
Solids Content							
Viscosity/Consistency			X	X	X	X	
Drying/ Cure Time	X	X	X	X	X	X	X
Application/Spray Properties	X						
Pot Life	X						
Adhesion[1]	X						
Gloss			X	X	X	X	
Color	X						
Appearance of Applied Coating	X	X	X	X	X	X	X
Mixing	X	X	X	X	X	X	X
Self-Lifting							

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TABLE I. Tests for FSC 8010 Paint, Dope, Varnish, and Related Products – Continued.

CHARACTERISTICS	SPECIFICATION						
	MPI 138	MPI 141	MPI 15RR	MPI 2	MPI 27	MPI 31	MPI 32
Condition in Container	X	X	X	X	X	X	X
Solids Content							
Viscosity/Consistency	X	X	X		X		X
Drying/ Cure Time	X	X	X	X	X	X	X
Application/Spray Properties							
Pot Life							
Adhesion[1]							
Gloss	X	X				X	
Color							
Appearance of Applied Coating	X	X	X	X	X	X	X
Mixing	X	X	X	X	X	X	X
Self-Lifting							

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TABLE I. Tests for FSC 8010 Paint, Dope, Varnish, and Related Products – Continued.

CHARACTERISTICS	SPECIFICATION							
	MPI 94	MPI 99	TT-C-492	TT-P-1757	TT-P-1952	TT-P-2760	TT-P-28	TT-P-645
Condition in Container	X	X	X	X	X	X	X	X
Solids Content								
Viscosity/Consistency	X		X	X	X	X	X	X
Drying/ Cure Time	X	X	X	X	X	X	X	X
Application/Spray Properties			X	X	X	X	X	X
Pot Life						X		
Adhesion[1]				X		X	X	X
Gloss	X	X		X				X
Color			X	X	X	X	X	X
Appearance of Applied Coating	X	X	X	X	X	X	X	X
Mixing	X	X	X	X	X	X	X	X
Self-Lifting								

NOTES:

Asterisk (*) next to the specification denotes inactive for new designs.

For a commodity not listed in TABLES I-V, see guidance on packaged product shelf-life extension at https://www.shelflife.dla.mil/. Program POCs for the Military Services and Agency Administrators are listed under the POCs tab.

Only an approved laboratory under the DoD Shelf-Life Program can authorize the customer to extend the shelf-life of any item.

See 5.10.1 for testing frequency and extension periods.

Certain packaging options do not readily allow for all tests to be completed such as aerosol cans, touch-up kits, and other self-contained kits. These types are exempt from solids content, viscosity/consistency, and pot life. All other tests indicated in Table I should be conducted as packaging allows.

Powder coatings that are formulation options in specifications are exempt from tests that are not physically possible due to the nature of the powder. All other tests indicated in Table I should be conducted as the powder formulation allows.

1. Adhesion refers to multiple methods, examples are wet/dry adhesion and knife test. When multiple tests for adhesion are listed wet adhesion and dry adhesion are satisfactory.

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TABLE II. Tests for FSC 8010 Chemical Agent Resistant Coating.

CHARACTERISTICS	SPECIFICATION			
	MIL-DTL-53039	MIL-DTL-53084	MIL-DTL-64159	MIL-PRF-32348
Condition in Container	X	X	X	X
Solids Content				
Viscosity/Consistency	X	X	X	
Drying/ Cure Time	X	X	X	
Application/Spray Properties	X	X	X	X
Pot Life			X	
Adhesion[1]	X	X	X	X
Gloss	X	X	X	X
Color	X	X	X	X
Appearance of Applied Coating	X	X	X	X
Mixing	X	X	X	
Self-Lifting				

NOTES:

For a commodity not listed in TABLES I-V, see guidance on packaged product shelf-life extension at https://www.shelflife.dla.mil/. Program POCs for the Military Services and Agency Administrators are listed under the POCs tab.

Only an approved laboratory under the DoD Shelf-Life Program can authorize the customer to extend the shelf-life of item.

Products are extendable once for 6 months.

Certain packaging options do not readily allow for all tests to be completed such as aerosol cans, touch-up kits, and other self-contained kits. These types are exempt from solids content, viscosity/consistency, and pot life. All other tests indicated in Table I should be conducted as packaging allows.

Powder coatings that are formulation options in specifications are exempt from tests that are not physically possible due to the nature of the powder. All other tests indicated in Table I should be conducted as the powder formulation allows.

1. Adhesion refers to multiple methods, examples are wet/dry adhesion and knife test. When multiple tests for adhesion are listed wet adhesion and dry adhesion are satisfactory.

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TABLE III. Tests for FSC 8010 Thinner and Remover.

CHARACTERISTICS	SPECIFICATION						
	A-A-3007	A-A-857	MIL-PRF-83936	*MIL-R-81294	MIL-DTL-81772	TT-R-248	TT-R-2918
Condition in Container	X	X	X	X	X	X	X
Flash Point	X				X		
Acidity	X	X					
Heptane Miscibility	X	X					
Spot Test		X			X		
Viscosity/Consistency			X	X		X	X
Flammability			X	X		X	
Corrosion			X				X
Paint Removal			X				
Effect on Metal (single panel)				X		X	
pH							X

NOTES:

Asterisk (*) next to the specification denotes inactive for new designs.

For a commodity not listed in TABLES I-V, see guidance on packaged product shelf-life extension at https://www.shelflife.dla.mil/. Program POCs for the Military Services and Agency Administrators are listed under the POCs tab.

Only an approved laboratory under the DoD Shelf-Life Program can authorize the customer to extend the shelf-life of any item.

See 5.10.1 for testing frequency and extension periods.

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TABLE IV. Tests for FSC 8030 Preservative and Sealing Compounds

CHARACTERISTICS	SPECIFICATION				
	A-A-272	A-A-52408	A-A-59293	ASTM-D5363	*MIL-C-10578
Condition in Container	X	X	X	X	X
Flow					
Application Life[1]	X				
Tack-Free Time	X				
Solids/Non-volatile Content					
Viscosity		X			
Drying/ Cure Time		X			
Hardness/Standard Cure Rate	X		X		
Flash Point					X
Corrosion Resistance					
Application Properties					
Pot Life/Working Life			X		
Adhesion			X		
Tensile Strength					
Strength at Standard Conditions				X	
Water Resistance					
Compressive Strength					
Shrinkage					
Acid Content					X
Flexibility					

TABLE IV. Tests for FSC 8030 Preservative and Sealing Compounds – Continued.

CHARACTERISTICS	SPECIFICATION					
	MIL-C-8514	MIL-DTL-450	MIL-DTL-81706	MIL-M-24041	MIL-PRF-16173	MIL-DTL-85054
Condition in Container	X	X	X	X	X	X
Flow						
Application Life[1]				X		
Tack-Free Time						
Solids/Non-volatile Content						
Viscosity	X	X		X		
Drying/ Cure Time	X	X				X
Hardness/Standard Cure Rate				X		
Flash Point					X	
Corrosion Resistance			X			X
Application Properties		X			X	
Pot Life/Working Life						
Adhesion			X			
Tensile Strength						
Strength at Standard						
Water Resistance		X				
Compressive Strength						
Shrinkage						
Acid Content						
Flexibility						

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TABLE IV. Tests for FSC 8030 Preservative and Sealing Compounds – Continued.

CHARACTERISTICS	SPECIFICATION					
	MIL-PRF-19565	MIL-PRF-23586	MIL-PRF-24176	*MIL-PRF-3043	*MIL-PRF-32171	MIL-PRF-6799
Condition in Container	X	X	X	X	X	X
Flow						
Application Life[1]						
Tack-Free Time						
Solids/Non-volatile Content						X
Viscosity		X		X	X	X
Drying/ Cure Time			X	X	X	X
Hardness/Standard Cure		X	X			
Flash Point						
Corrosion Resistance						
Application Properties					X	
Pot Life/Working Life			X		X	
Adhesion						
Tensile Strength						X
Strength at Standard						
Water Resistance	X					
Compressive Strength			X			
Shrinkage			X			
Acid Content						
Flexibility	X					

TABLE IV. Tests for FSC 8030 Preservative and Sealing Compounds – Continued

CHARACTERISTICS	SPECIFICATION						
	MIL-PRF-8116	MIL-PRF-81733	*MIL-PRF-8516	*MIL-R-9300	MIL-S-11030	MIL-S-15204	MIL-PRF-81309
Condition in Container	X	X	X	X	X	X	X
Flow		X					
Application Life[1]		X	X				
Tack-Free Time		X					
Solids/Non-volatile	X				X	X	X
Viscosity	X				X	X	
Drying/ Cure Time							
Hardness/Standard Cure		X	X				
Flash Point						X	X
Corrosion Resistance							X
Application Properties							X
Pot Life/Working Life							
Adhesion					X		
Tensile Strength				X			
Strength at Standard							
Water Resistance							
Compressive Strength							
Shrinkage							
Acid Content							
Flexibility							

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TABLE IV. Tests for FSC 8030 Preservative and Sealing Compounds – Continued.

CHARACTERISTICS	SPECIFICATION					
	MIL-S-45180	*MIL-S-85334	SAE-AMS3265	SAE-AMS3276	SAE-AMS3277	SAE-AMS3284
Condition in Container	X	X	X	X	X	X
Flow		X	X	X	X	
Application Life[1]			X	X	X	X
Tack-Free Time			X	X	X	X
Solids/Non-volatile Content	X	X				
Viscosity	X					X
Drying/ Cure Time						
Hardness/Standard Cure Rate			X	X	X	X
Flash Point						
Corrosion Resistance						
Application Properties						
Pot Life/Working Life						
Adhesion						
Tensile Strength						
Strength at Standard Conditions						
Water Resistance						
Compressive Strength						
Shrinkage						
Acid Content						
Flexibility						

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TABLE IV. Tests for FSC 8030 Preservative and Sealing Compounds – Continued.

CHARACTERISTICS	SPECIFICATION			
	SAE-AMS3374	SAE-AMS-S-4383	SAE-AMS-S-7124	SAE-AMS-S-8802
Condition in Container	X	X	X	X
Flow				X
Application Life[1]	X		X	X
Tack-Free Time	X	X		X
Solids/Non-volatile Content				
Viscosity		X	X	X
Drying/ Cure Time				
Hardness/Standard Cure Rate	X		X	X
Flash Point				
Corrosion Resistance				
Application Properties				
Pot Life/Working Life				
Adhesion				
Tensile Strength				
Strength at Standard				
Water Resistance				
Compressive Strength				
Shrinkage				
Acid Content				
Flexibility				

NOTES:

Asterisk (*) next to the specification denotes inactive for new designs.

For a commodity not listed in TABLES I-V, see guidance on packaged product shelf-life extension at https://www.shelflife.dla.mil/. Program POCs for the Military Services and Agency Administrators are listed under the POCs tab.

Only an approved laboratory under the DoD Shelf-Life Program can authorize the customer to extend the shelf life of any item.

See 5.10.1 for testing frequency and extension periods.

1. Application life may also be called application time or extrudability.

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TABLE V. Tests for FSC 8040 Adhesives and Related Products.

CHARACTERISTICS	SPECIFICATION					
	A-A-3097	A-A-342	A-A-50535	MIL-A-21016	MIL-A-24179	MIL-A-24456
Condition in Container	X	X	X	X	X	X
Bond Strength		X				
Viscosity				X	X	X
Drying/ Cure Time		X				
Hardness						X
Lap Shear Strength						
Application Properties						
Flow						
Extrusion						
Peel Strength						
Tensile Strength	X				X	
Adhesive Strength/Adhesion				X		
Pot Life						X
Tack Free						
pH			X			
Flexibility						

TABLE V. Tests for FSC 8040 Adhesives and Related Products – Continued.

CHARACTERISTICS	SPECIFICATION					
	MIL-DTL-3316	MIL-A-46106	MIL-A-46146	MIL-A-5540	*MIL-A-81236	MIL-PRF-9117
Condition in Container	X	X	X	X	X	X
Bond Strength						X
Viscosity	X	X	X	X		
Drying/ Cure Time	X					
Hardness		X	X		X	
Lap Shear Strength					X	
Application Properties						
Flow		X	X			
Extrusion		X	X			
Peel Strength		X	X			X
Tensile Strength						
Adhesive Strength/Adhesion	X			X		
Pot Life				X		
Tack Free		X	X			
pH						
Flexibility	X					

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TABLE V. Tests for FSC 8040 Adhesives and Related Products – Continued.

CHARACTERISTICS	SPECIFICATION					
	MMM-A-121	MMM-A-132	MMM-A-134	MMM-A-1617	MMM-A-1754	MMM-A-1058
Condition in Container	X	X	X	X	X	X
Bond Strength						
Viscosity	X					
Drying/ Cure Time						
Hardness						
Lap Shear Strength		X	X		X	
Application Properties						X
Flow					X	
Extrusion						
Peel Strength						
Tensile Strength						
Adhesive Strength/Adhesion	X			X		
Pot Life					X	
Tack Free						
pH						
Flexibility						

NOTES:

Asterisk (*) near the specification denotes inactive for new designs.

For a commodity not listed in TABLES I-V, see guidance on packaged product shelf-life extension at https://www.shelflife.dla.mil/. Program POCs for the Military Services and Agency Administrators are listed under the POCs tab.

Only an approved laboratory under the DoD Shelf-Life Program can authorize the customer to extend the shelf life of any item.

See 5.10.1 for testing frequency and extension periods.

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APPENDIX A

ACRONYMS AND INITIALISMS

A.1 SCOPE.

A.1.1 Scope. This appendix is not a mandatory part of this Standard. The information contained herein is intended for guidance only. This appendix lists most of the common acronyms and initialisms used in this document.

AMSC	Acquisition Management Systems Control
ASSIST	Acquisition Streamlining and Standardization Information System
CID	Commercial Item Descriptions
DLA	Defense Logistics Agency
DLIS	Defense Logistics Information Service
DoD	Department of Defense
EA	Executive Agency
FSC	Federal Supply Class
ISO	International Organization for Standardization
NGS	Non-Government Standards
NSN	National Stock Number
OSHA	Occupational Safety and Health Administration
PQDR	Product Quality Deficiency Report
QA	Quality Assurance
QAR	Quality Assurance Representative
QAS	Quality Assurance Specialist
QC	Quality Control
QPD	Qualified Products Database
QPL	Qualified Products List
QS	Quality Surveillance
QSL	Quality Surveillance List
SDS	Safety Data Sheet
USG	U.S. Gallons

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APPENDIX B

MILITARY SERVICE LABORATORIES AND TESTING CAPABILITIES

B.1 SCOPE.

B.1.1 Scope. This appendix is not a mandatory part of this Standard. The information contained herein is intended for guidance only. Unless otherwise directed, all samples should be forwarded to the laboratory designated in the appropriate service regulations. It is recommended to contact the laboratories listed below prior to submission of samples to ensure capabilities exist to perform/accomplish sample testing requirements.

B.2 LIST OF LABORATORIES.

B.2.1 Laboratories. For a list of approved laboratories, see the table of QSL Test Labs for chemical commodities at https://www.shelflife.dla.mil/site/sles/TestLabs.aspx, Common Access Card (CAC) enabled.

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CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – AS
DLA – CQ

Review activities:
Army – CR, MI
Navy – MC
DLA – PS

Preparing activity:
DLA – CQ
(Project 80GP-2021-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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